

Task: Automated Trading Bot Research + Paper Trading Prototype

Goal

Create an automated trading system/bot that executes trades on a **paper trading account** first, with architecture that can later scale to a **real trading account**.

The bot must only work with these US stock tickers:

AMZN, TSLA, MSFT, AAPL, META, NVDA, GOOGL, JPM

What You Need To Do

1. Research

Find and study relevant/new companies, platforms, APIs, and tools that are building or enabling automated stock trading bots.

You need to identify:

- What platforms already exist
- Which APIs allow automated stock trading
- Which ones support paper trading
- Which ones can later support real-money trading
- What limitations they have
- Whether they allow the frequency of trading required by our bot

Do not just list companies. Explain what we can learn from each one.

2. Choose Execution Platform

Find the best platform/library/API for:

- Paper trading
- Real trading later
- US stocks
- Automated order execution
- Reliable historical data
- Safe transition from paper to live account

You must clearly explain why you selected it.

3. Build Trading Strategy Logic

Come up with a real-world trading strategy for the bot.

The strategy can be anything, but it must be practically executable using available APIs/platforms.

You must clearly define:

- When the bot buys
- When the bot sells
- How often it checks the market
- How much capital it allocates per stock
- Risk limits
- Stop-loss rules
- Profit-taking rules
- Maximum number of open positions
- What happens if the API fails
- What happens if market data is missing

The strategy must only trade:

AMZN, TSLA, MSFT, AAPL, META, NVDA, GOOGL, JPM

4. Build Paper Trading Bot

Create a working prototype that:

- Connects to a paper trading account
- Fetches market data
- Runs the strategy automatically
- Places buy/sell orders
- Tracks portfolio value
- Logs every trade
- Stores every decision made by the bot
- Shows current holdings and profit/loss

The bot must not use real money in MVP.

5. Backtesting

You must prove whether the strategy works using historical data.

Run the same strategy over these periods:

- 1 week
- 1 month
- 3 months
- 6 months
- 1 year
- 3 years

For each period, show:

- Starting capital
- Ending capital
- Total profit/loss
- Percentage return
- Number of trades
- Winning trades
- Losing trades
- Max drawdown
- Fees/slippage assumptions
- Full trade log

6. Proof Requirement

The final submission must prove whether the bot made profit or loss in each period.

You must not cherry-pick trades.

You must not manually modify results.

The same strategy logic must be used across all timeframes.

7. Final Deliverables

Submit:

1. Research document
2. Selected platform/API explanation
3. Strategy document
4. Working paper trading bot
5. Backtesting report
6. Trade logs
7. Profit/loss summary for all required periods
8. Instructions to run the bot
9. Clear explanation of what is needed before using real money

Success Criteria

This task is complete only when:

- The bot can run on a paper trading account
- The bot only trades the approved 8 tickers
- The strategy is clearly explained
- The backtest is reproducible
- Profit/loss is shown for all 6 periods
- Every trade is logged
- The system can later be connected to a real account with minimal changes